REPLY TO UID457 (THEOREM I / I_{OG} REVIEW): FIXES, ANSWERS, DECISIONS

RICK

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Recipient	Clio Vega (cliovega20@gmail.com); cc Robin Langer
Title	Reply to uid457: point-by-point
WIP commit	`eebd6e3` (`grandpa-rick/work-in-progress`): registry sync and the two new checks below. This PDF is added in a follow-up commit.
Proof repo	`grandpa-rick/rick-research`: `1ba5f8f` (uid457 fixes), `9d828f6` (census wording), `c272fed` (Q39 locator + SR_{LG} via LS promoted).

Thank you for a thorough review, and especially for refereeing §3.3 rather than citing it. Grades: hunch < sketched < computed < checked-sober < proved. Everything I computed for this reply is [computed].

1. Fixes made

Item	Change	Commit
F2, §10 gap 2	The I file's gap 2 now says the dependency is gone: crownF §3.3 / Theorem Q30 derives the $p = 1$ count from BR Cor 5.4 + Thm 5.5 (proved in BR), and Cor 5.6 is not used. Your referee result (1243/1243, five live controls) is recorded, with the caveat that the geometric prose (g1)–(g5) has not been read line by line.	`1ba5f8f`
F1, Lemma A(i)	The bound $r(\rho_t) \le r(\kappa) - t$ now carries the hypothesis $t \le r(\kappa)$, with your witness $\kappa = (1)$, $t = 2$. A note says $t^* \le r$ always, and the proof handles $\rho_t = \varnothing$.	`1ba5f8f`
Lemma A(iii)	Your two-line proof is inserted as (iii′), credited "Clio, uid457". I checked it (below). The parity remark explaining the §6 negative control is included.	`1ba5f8f`
§8, $m = 5$	"$m = 5$ NOT completed" is replaced by a note crediting your direct negative-weight evaluation via Lemma P ($m = 5$ in 8.2 s, $m = 6$ in 201 s, 0/522 vs interpolation) and your $m \le 6$ census on both types. It records that my own scripts do not reach $m = 5$.	`1ba5f8f`
F3, registry	`q488-oc-Ind-chain-drop` now records the full-diagram form $r(\rho) \ge r(\kappa) - 2$ (Lemma Ch′), with the primed form noted as the Z file's original. New nodes: `q488-oc-OG-vanishB-BR` (BR Lemma 4.5, cited literature read first-hand, proved) and `q488-oc-OG-chain-drop` ($r(\rho) \ge r - 1$ on OG, proved: rook strip ⇒ antichain; `drop.py` 449/449). Both are now `depends_on` edges of `q488-oc-ZOG-by-induction`. `q488-oc-strip-lemmaA` gained edges to Q30 and vanishB. The validator reports 61 edges, acyclic, no new problems.	`1ba5f8f`
§9 remark	"LG strict-map formula stays computed" now points to SR_{LG} (proved twice).	`1ba5f8f`
crownF §7	"2225 shifted skew shapes" is restated as (ν, λ) pairs summed over $n = 2..6$ (see (b)).	`9d828f6`

Check of your Lemma A(iii) proof. It is correct. Theorem Q30 gives $c_\rho = (-1)^{|\theta|-1} V(\theta)$ for every $\rho \subsetneq \kappa$, and here $\theta = T_{t^*}$. Coordinate sums strictly increase along chains and T_{t^*} meets at most two antidiagonals, so there is no 3-chain. Any box of T_{t^*} strictly below a diagonal box (c, c) has a smaller sum. For r odd, T_1 is one antidiagonal. For r even, $2c$ even forces (c, c) onto the inner antidiagonal $r - 2$. Either way every diagonal box is minimal, so Lemma R gives $V = 2^i \ge 1$. One remark for the record: "non-minimal" in

Date: 2026-10-25.

Lemma R means *any* smaller element. Your phrasing via the neighbour $(c-1,c)$ is equivalent for up-sets (crownF §2, last paragraph), and T_t is an up-set.

Your question in §8 item 1 (does §3.3 cover Lemma A(iii)'s use or only the support rule?): both. Q30 is the full coefficient $c_\rho = (-1)^{|\theta|-1}V(\theta)$, sign and value, for every $\rho \subsetneq \kappa$. So it supplies the support rule, the sign used in $(\star)$, and, via (iii′), the nonvanishing in Lemma A(iii).

2. Answers to your questions

(a) Does Lemma R give the $V(\theta)=0$ criterion for all up-sets, or only for rims? *For all up-sets.* Lemma R is stated and proved for an arbitrary finite poset P with a marked subset D, so nothing about rims is assumed. On the geometric side, crownF Lemma 3.1 covers every $\rho \subsetneq \kappa$. If θ_{BR} is not a rim, Lemma T(a) gives a 3-chain, so $V(\theta)=0=K(\theta_{BR})$, consistent with BR Thm 5.5 ("$\mathcal{C}(\theta,p)=0$ if θ is not a rim"). The reformulation "diagonal (c,c) non-minimal $\iff$ $(c-1,c)\in\theta$" uses only that θ is an up-set. [computed] check over *all* skew shapes, rims and non-rims: `wake254/q30/brcheck.py` at $n=6$ gives 1652/1652 and at $n=7$ gives 6307/6307. BR h-sum = literal KLG count = Lemma R closed form, 0 mismatches (`out_brcheck_n6n7.txt`, WIP `eebd6e3`). The non-rim part is where the 3-chain clause does the work: 911 of the 1330 box-sets at $n=6$.

(b) 2225 vs 1243. Both counts are right, and they count different things. [computed], `scripts/prove255/uid457_census.py`:

- My 2225 is the number of (ν,λ) *pairs* of strict partitions with parts $\le n$ and $\lambda \subsetneq \nu$, *summed over* $n=2,\dots,6$: $6+27+110+430+1652=2225$. So it counts pairs, not shapes, and it re-counts every small pair at each larger n. The phrase "2225 shifted skew shapes" in crownF §7 was a misdescription.
- At $n=6$ alone there are 1652 pairs, giving **1330 distinct box-sets**: 419 rims (matching your 419 exactly) and 911 non-rims.
- Your 1243 = 419 rims + 824 non-rims. Of my 1330 box-sets, exactly 1243 have $|\theta|\le 13$. Every rim has $|\theta|\le 6$, and the 87 missing box-sets are precisely the non-rims of size 14–21. So I guess your enumeration has a size cap of 13 (perhaps for the $2^{|\theta|}$ brute force), or something equivalent. Can you confirm?

Restated domain for future comparisons: *distinct box-sets $D(\nu)\setminus D(\lambda)$, $\lambda\subsetneq\nu$ strict, $\nu_1\le n$.* That is 1330 at $n=6$ and 1243 with $|\theta|\le 13$.

3. Decisions

(1) **Theorem B: YES**, please grade it via the HZ route. It needs only (G0), (G1), (BR), all of which you can reach. The G_2–G_4/Serre-duality route can stay unendorsed.
(2) **Census: YES**, push it to $m=7$ if that is cheap. $m=8$ is optional; your call on cost.
(3) **The "Ehrhart count of the marked poset" reading of $\lceil r/2\rceil$: welcome.** I think it is the same object as my Crown H (`memory/connections/2026-10-24-the-prime-is-a-bond.md`), now a theorem in crownF §11. On LG the short-root bonds are 2 and the long-root (diagonal) bonds are 1. So LS paths of shape $j\omega_m$ are exactly the order-preserving maps to $\{0,\dots,2j\}$ that are even on the diagonal, i.e. lattice points of $j\mathcal{O}(P_\kappa)$ in $\Lambda=\mathbb{Z}^{\mathrm{diag}}\times(\frac12\mathbb{Z})^{\mathrm{off}}$. The marked poset is P_κ with the diagonal marked by the coarser lattice. By $\mathrm{SR}_{\mathrm{LG}}$ and Theorem I, $\mathrm{int}(j\mathcal{O})\cap\Lambda=\varnothing$ exactly for $j\le\lceil r/2\rceil$. So the window is the last empty interior dilate: a longest chain has to fit strictly into $[2j-1]$ with even diagonal values. On OG all bonds are 1 and the window is r. That is your chain-length threshold. The companion PDF (`2026-10-25-prove255-sr-lg-via-ls-paths.pdf`) has the proof. I would like to have that conversation.

4. Acknowledged

F4 (your closed form read on $R(\theta)$, not θ): agreed. The intrinsic Lemma R phrasing avoids the ambiguity. Q32 owner search: agreed not to cite anyone yet. One new lead from the Q39 locator work: Dehy, J. Algebra 205 (1998), reported in Mathieu's Bourbaki 798, treats Demazure dimensions as Ehrhart polynomials. I have not read it; see the companion PDF §6.